Redlining in the Bay Area

Emily Ma Larissa Arreola Moreno Vittal Vasudevan
Daniel Porter

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1 Redlining in the Bay Area

1.1 Exploration and processing of data

1.1.1 Importing libraries and data

```
import pandas as pd
import geopandas as gpd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import matplotlib.patches as mpatches
from src.holc_utils import *

sns.set(style="whitegrid", context="notebook")
sns.set_palette('rocket')
fig_folder = "figures/redlining_bay_area/"
```

We are now importing the main dataset we obtained from Mapping Inequality (<this is a url), and processed in the script `data_preprocessing.py`.

```
# loading HOLC data
holc = get_data_file("data/holc_bayarea.json")
holc.head()
```

	area_id	city	state	city_survey	category	grade	label	residential	commercial	industrial	...	GEOIDFQ	GEOID	NAME	NAMELSAD	STUSPS	STATE_NAME	LSAD	ALAND	AWATER	geometry
0	6690	Oakland	CA	True	Best	A	A	True	False	False	...	0500000US06001	06001	Alameda	Alameda County	CA	California	06	[illegible]	[illegible]	POLYGON (((-122.20252 37.81099, -122.2021...
1	6690	Oakland	CA	True	Best	A	A	True	False	False	...	0500000US06013	06013	Contra Costa	Contra Costa County	CA	California	06	[illegible]	[illegible]	POLYGON (((-122.20252 37.81099, -122.2021...
2	6662	Oakland	CA	True	Best	A	A1	True	False	False	...	0500000US06001	06001	Alameda	Alameda County	CA	California	06	[illegible]	[illegible]	POLYGON (((-122.25734 37.89463, -122.2580...
3	6662	Oakland	CA	True	Best	A	A1	True	False	False	...	0500000US06013	06013	Contra Costa	Contra Costa County	CA	California	06	[illegible]	[illegible]	POLYGON (((-122.25734 37.89463, -122.2580...
4	6710	Oakland	CA	True	Best	A	A10	True	False	False	...	0500000US06001	06001	Alameda	Alameda County	CA	California	06	[illegible]	[illegible]	POLYGON (((-122.23403 37.80671, -122.2350...

5 rows × 24 columns

In order for us to map the geographic data, we will also require Bay Area's counties outline.

```
# make sure we have the same coordinate reference system (crs)
correct_crs = holc.crs
# get bay outline
bay_outline = get_bay_counties(correct_crs)
```

```
holc.columns
```

```
Index(['area_id', 'city', 'state', 'city_survey', 'category', 'grade', 'label',
       'residential', 'commercial', 'industrial', 'fill', 'STATEFP',
       'COUNTYFP', 'COUNTYNS', 'GEOIDFQ', 'GEOID', 'NAME', 'NAMELSAD',
```

```
       'STUSPS', 'STATE_NAME', 'LSAD', 'ALAND', 'AWATER', 'geometry'],
      dtype='object')
```

Since we are going to work with the grades of areas, we need to get to know possible missing values or errors in our data. First we'll explore if there are areas without a grade assigned.

```
print(f"Grades assigned:\n{holc.grade.unique()}")
print(f"Categories assigned:\n{holc.category.unique()}")
```

```
Grades assigned:
['A' 'B' 'C' 'D' None]
Categories assigned:
['Best' 'Still Desirable' 'Definitely Declining' 'Hazardous'
 'Industrial and Commercial ' 'Industrial' 'Commercial']
```

As we can see, there are 5 possible values in the "grade" column, including `None`, let's see what is happening to those with `None` assigned.

```
holc[holc["grade"].isna()]
```

area_id	city_id	city	state	city_survey	category	grade	label	residential	commercial	industrial	...	GEO_ID	NAME	NAMELSAD	STATE_NAME	[illegible]	geometry
127	111172	Oakland	CA	True	Industrial and Commercial	None	Industrial and Commercial	False	True	True	...	0500000US06001	Alameda	Alameda County	California	[illegible]	MULTIPOLYGON (((-122.15608 37.72594, -122.1573...
229	11138	San Francisco	CA	True	Industrial and Commercial	None	Industrial and Commercial	False	True	True	...	0500000US06081	San Mateo	San Mateo County	California	[illegible]	MULTIPOLYGON (((-122.39492 37.72951, -122.3951...
230	11138	San Francisco	CA	True	Industrial and Commercial	None	Industrial and Commercial	False	True	True	...	0500000US06075	San Francisco	San Francisco County	California	[illegible]	MULTIPOLYGON (((-122.39492 37.72951, -122.3951...
268	11149	San Jose	CA	True	Industrial	None	Industrial	False	False	True	...	0500000US06085	Santa Clara	Santa Clara County	California	[illegible]	MULTIPOLYGON (((-121.87293 37.31664, -121.8751...
269	11150	San Jose	CA	True	Commercial	None	Commercial	False	True	False	...	0500000US06085	Santa Clara	Santa Clara County	California	[illegible]	MULTIPOLYGON (((-121.89659 37.30408, -121.8975...

5 rows × 24 columns

As we can see, there are only 5 rows with a `None` grade, which also matches those that are categorized as commercial and/or industrial areas, just to make sure it is a 1:1 relationship, let's see if there are industrial and/or commercial areas that do have a grade assigned to them.

```
holc[(holc["commercial"]) | (holc["industrial"])]
```

area…	city	state	city…	category	grade	label	resid…	comm…	indust…	…	GEO_ID	…	NAME	…	STATE_NAME	…	geometry
127 11172	Oakland	CA	True	Industrial and Commercial	None	Industrial and Commercial	False	True	True	…	0500000US06001	06 001	Alameda County	[illegible]	California	[illegible]	POLYGON (((-122.15608 37.72594, -122.1573...
229 11138	San Francisco	CA	True	Industrial and Commercial	None	Industrial and Commercial	False	True	True	…	0500000US06081	06 081	San Mateo County	[illegible]	California	[illegible]	POLYGON (((-122.39492 37.72951, -122.3951...
230 11138	San Francisco	CA	True	Industrial and Commercial	None	Industrial and Commercial	False	True	True	…	0500000US06075	06 075	San Francisco County	[illegible]	California	[illegible]	POLYGON (((-122.39492 37.72951, -122.3951...
268 11149	San Jose	CA	True	Industrial	None	Industrial	False	False	True	…	0500000US06085	06 085	Santa Clara County	[illegible]	California	[illegible]	POLYGON (((-121.87293 37.31664, -121.8751...
269 11150	San Jose	CA	True	Commercial	None	Commercial	False	True	False	…	0500000US06085	06 085	Santa Clara County	[illegible]	California	[illegible]	POLYGON (((-121.89659 37.30408, -121.8975...

5 rows × 24 columns

As we can see, those are the same 5 columns as before, so we are now sure that only the commercial and/or industrial categories are the only ones without a grade assigned. Since we are interested in the dataset as a whole, we deem valuable to keep those rows, but we will modify the grade value so that we can work with it later.

We are also going to make an additional column combining category and grades, since they are closely related, and it would give us a better understanding of the historical perspectives.

```
# unify commercial/industrial category
conditions = [(holc["category"] == 'Industrial and Commercial ')|
              (holc["category"] == 'Industrial')|
              (holc["category"] == 'Commercial')]
```

```
choices = ['Commercial and/or Industrial']
holc["category"] = np.select(conditions, choices,default=holc["category"])

# imputing values
holc['grade'] = holc['grade'].fillna("Comm/Ind")
# adding grade -category column
holc["grade_category"] = holc["grade"] + " - " + holc["category"]
```

Lastly we will add an area column, for each of the polygons that we have in our geometries, so that we can use area as another aspect of our analysis.

```
# convert data to a projection usable to calculate areas
holc["area_km2"] = holc.to_crs(epsg=3857).area / 1e6
```

1.1.2 Analysis

In this section we will try to answer the following questions:

1. How many HOLC-graded areas exist in the Bay Area?
2. What distribution of areas were graded A, B, C, and D?
3. How much land area was assigned each HOLC grade?
4. Were certain grades more geographically extensive?

All of these while comparing across cities.

But first, to be able to plot according to the colors on the data set, we will extract them and store them in a dictionary.

```
grade_color_map = (holc[["grade_category", "fill"]].drop_duplicates().set_index("grade_categ
grade_color_map
```

```
{'A - Best': '#76a865',
 'B - Still Desirable': '#7cb5bd',
 'C - Definitely Declining': '#ffff00',
 'D - Hazardous': '#d9838d',
 'Comm/Ind - Commercial and/or Industrial': '#000000'}
```

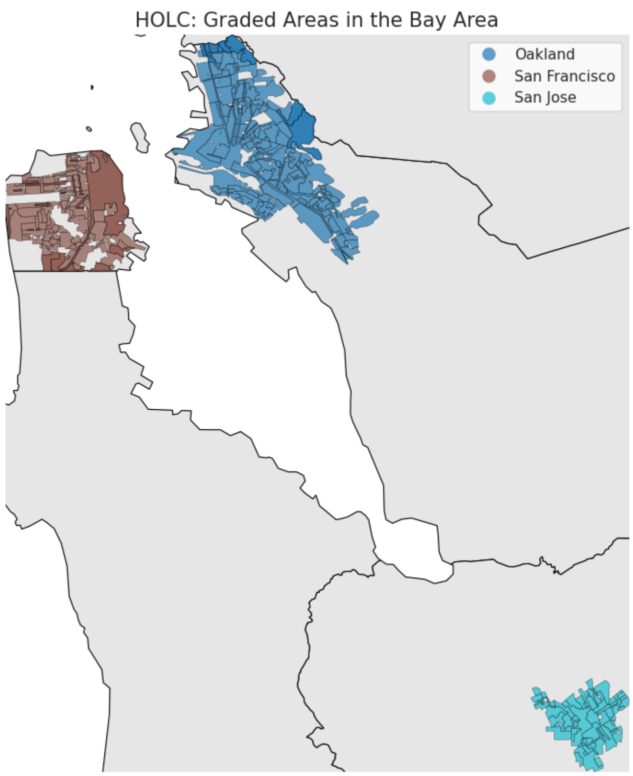
Question 1: How many HOLC-graded areas exist in the Bay Area?

```
# subsetting data - grouping by cities and counties
print(holc.groupby(["city","NAMELSAD"]).size().reset_index(name="count_holc_areas"))
city_counts = holc.groupby(["city"]).size().reset_index(name="count_holc_areas")
```

```
city              NAMELSAD  count_holc_areas
0        Oakland        Alameda County               120
1        Oakland   Contra Costa County                 8
2  San Francisco  San Francisco County                98
3  San Francisco      San Mateo County                 5
4       San Jose    Santa Clara County                39
```

From the table before, the only cities that we have in our dataset are Oakland, San Francisco and San Jose. However, probably due to the old mapping of land we can find both Oakland and San Francisco in two different counties each. This is just something to take note of but in reality we won't use that level of disaggregation, we just need the count per city. Let's take a look at the map.

```
fig_name_holc_graded = "1_holc_graded_areas.png"

fig, ax = plt.subplots(figsize=(10, 10))

# Bay area counties outline
bay_outline.plot(
    ax=ax,
    color="#e6e6e6",
    edgecolor="black",
    linewidth=0.8,
    alpha=1.0
)

# plot polygons of dataset
holc.plot(
    ax=ax,
    column="city",
    categorical=True,
    legend=True,
    alpha=0.7,
    edgecolor="black",
    linewidth=0.3
)

# bound map limits to the polygons we have
ax.set_xlim(holc.total_bounds[0], holc.total_bounds[2])
ax.set_ylim(holc.total_bounds[1], holc.total_bounds[3])

ax.set_title("HOLC: Graded Areas in the Bay Area", fontsize=15)
ax.set_axis_off()

fig.savefig(fig_folder + fig_name_holc_graded)
plt.show()
```



Now let's see how those graded cities are disaggregated to the level of grade/commercial or industrial category.

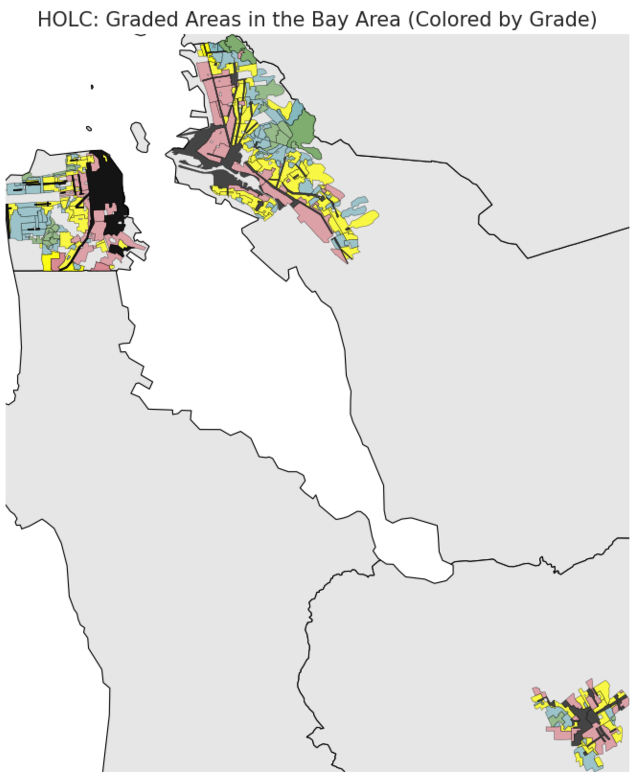
```
# aggregate data by city and grade (include grade+category for labeling)
grade_counts = holc.groupby(["city","grade","grade_category"]).size().reset_index(name="coun
# adding the count of total holc areas for that city, for clarity
grade_counts = grade_counts.merge(city_counts, on=["city"], how="left")
# selecting important columns
grade_counts = grade_counts[["city","grade","grade_category","count_graded_areas","count_hol
# get relative count of grades
grade_counts["relative_count"] = round(grade_counts["count_graded_areas"] / grade_counts["co
print(grade_counts)
```

```
city     grade                          grade_category  \
0         Oakland         A                              A - Best
1         Oakland         B                  B - Still Desirable
2         Oakland         C             C - Definitely Declining
3         Oakland  Comm/Ind  Comm/Ind - Commercial and/or Industrial
4         Oakland         D                         D - Hazardous
5   San Francisco         A                              A - Best
6   San Francisco         B                  B - Still Desirable
7   San Francisco         C             C - Definitely Declining
```

```
8   San Francisco  Comm/Ind  Comm/Ind - Commercial and/or Industrial
9   San Francisco         D                             D - Hazardous
10       San Jose         A                                  A - Best
11       San Jose         B                      B - Still Desirable
12       San Jose         C                C - Definitely Declining
13       San Jose  Comm/Ind  Comm/Ind - Commercial and/or Industrial
14       San Jose         D                             D - Hazardous

    count_graded_areas  count_holc_areas  relative_count
0                   16               128            12.0
1                   49               128            38.0
2                   42               128            33.0
3                    1               128             1.0
4                   20               128            16.0
5                   13               103            13.0
6                   36               103            35.0
7                   33               103            32.0
8                    2               103             2.0
9                   19               103            18.0
10                   2                39             5.0
11                   7                39            18.0
12                  17                39            44.0
13                   2                39             5.0
14                  11                39            28.0
```

```
fig_name_grade_city = "2_holc_grade_by_city.png"
fig, ax = plt.subplots(figsize=(10, 10))

bay_outline.plot(
    ax=ax,
    color="#e6e6e6",
    edgecolor="black",
    linewidth=0.8,
    alpha=1.0
)

holc.plot(
    ax=ax,
    color=holc["fill"],
    alpha=0.7,
    edgecolor="black",
    linewidth=0.3
)

ax.set_xlim(holc.total_bounds[0], holc.total_bounds[2])
ax.set_ylim(holc.total_bounds[1], holc.total_bounds[3])
```

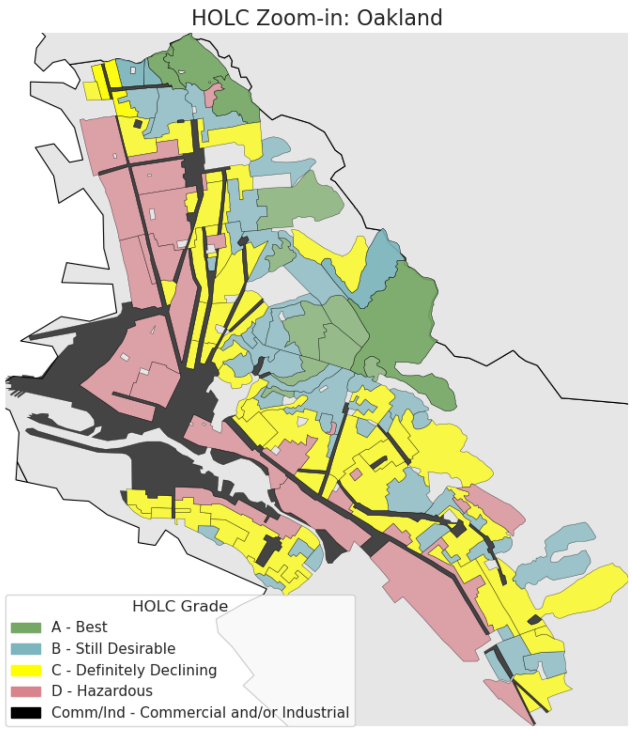
```
ax.set_title("HOLC: Graded Areas in the Bay Area", fontsize=15)
ax.set_axis_off()

fig.savefig(fig_folder + fig_name_grade_city)
plt.show()
```

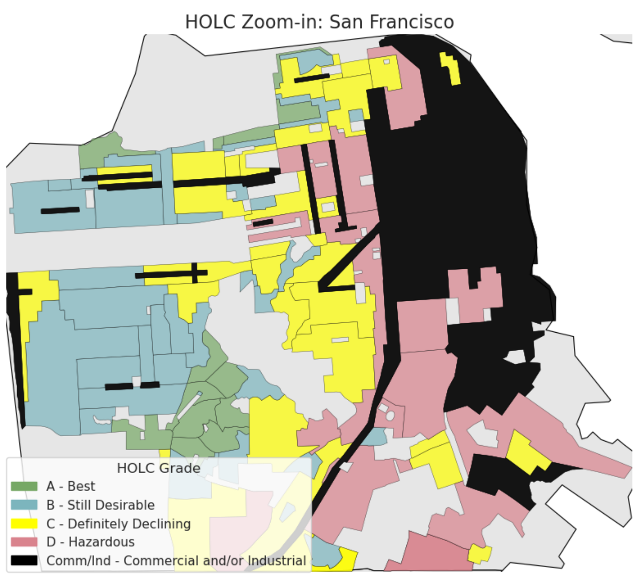


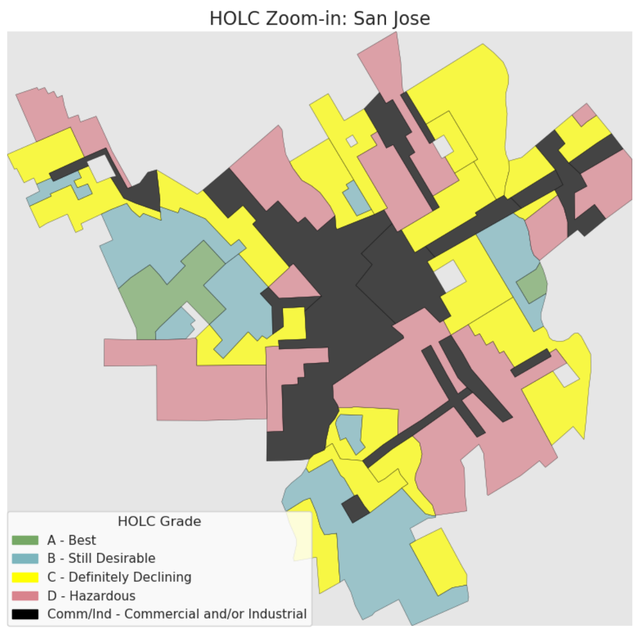
Let's take a closer look for each of the cities.

```
# List of unique HOLC cities
cities = holc["city"].unique()

for city in cities:

    fig_unique_city = f"3_unique_map_{city}.png"

    holc_city = holc[holc["city"] == city]

    fig, ax = plt.subplots(figsize=(8, 8))

    # bay area outline
    bay_outline.plot(
```

```
    ax=ax,
    color="#e6e6e6",
    edgecolor="black",
    linewidth=0.8,
    alpha=1.0
)

# just like before, plot the city and fill by grade
holc_city.plot(
    ax=ax,
    color=holc_city["fill"],
    alpha=0.7,
    edgecolor="black",
    linewidth=0.3
)

# zoom into city
bounds = holc_city.total_bounds
ax.set_xlim(bounds[0], bounds[2])
ax.set_ylim(bounds[1], bounds[3])

# format and legend
legend_handles = []
for grade, color in grade_color_map.items():
    patch = mpatches.Patch(color=color, label=f"{grade}")
    legend_handles.append(patch)

ax.legend(
    handles=legend_handles,
    title="HOLC Grade",
    loc="lower left",
    # bbox_to_anchor=(1.02, 1),
    borderaxespad=0
)

ax.set_title(f"HOLC Zoom -in: {city}", fontsize=16)
ax.set_axis_off()

fig.savefig(fig_folder + fig_unique_city, bbox_inches='tight')
plt.tight_layout()
plt.show()
```


HOLC Zoom-in: Oakland
HOLC Grade
A - Best
B - Still Desirable
C - Definitely Declining
D - Hazardous
Comm/Ind - Commercial and/or Industrial


HOLC Zoom-in: San Francisco
HOLC Grade
A - Best
B - Still Desirable
C - Definitely Declining
D - Hazardous
Comm/Ind - Commercial and/or Industrial

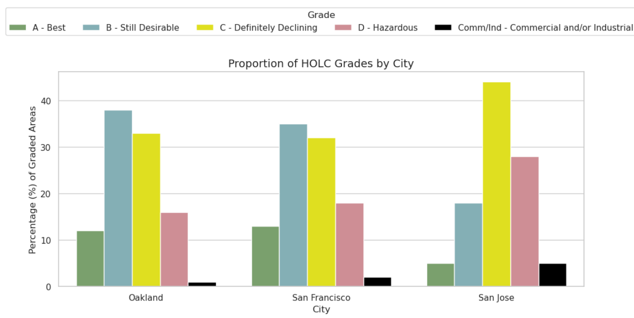


From the previous figure we can observe how the D (Hazardous, red) and C (Definitely declining, yellow) areas are surrounding the commercial and industrial areas (black), and how the A/B graded areas are not as spread out as the other grades, they are mainly concentrated in certain sections of the cities. Also notice how it seems the the A/B areas are less in each city, relative to C/D or even commercial and/or industrial in the case of San José.

Let's see now the proportion of HOLC grades by city, considering the total graded areas per city how many of those are in each grade or are commercial and/or industrial. As we can see, only in the case of San José this difference might be really evident, however to actually be able to conclude anything, we would have to carry out tests to prove our hypothesis.

```
fig_name_prop_bars = "4_proportion_grades_city.png"

fig, ax = plt.subplots(figsize=(12, 6))

sns.barplot(
    data=grade_counts,
    x="city",
    y="relative_count",
    hue="grade_category",
    palette=grade_color_map,
    hue_order=grade_color_map.keys()
)

plt.legend(title="Grade",loc='lower center',
```

```
            bbox_to_anchor=(0.5, 1.15),
            ncol=5)
plt.title("Proportion of HOLC Grades by City", fontsize=14)
plt.xlabel("City")
plt.ylabel("Percentage (%) of Graded Areas")
plt.tight_layout()
fig.savefig(fig_folder + fig_name_prop_bars)
plt.show()
```



Lastly, we are going to compare areas between grades, even though the proportion of graded polygons show us some information, it is also worth noting that the areas between polygons aren't equal, since HOLC was not a systematic and studied process, but rather subjective following discrimination more than anything else.

As an important note: the total area doesn't add up to 100% exactly, it is usually over a 100%, because of different historical reasons like overlap, the transition from physical to digital media, and the somewhat artisanal mapping, more prone to mistakes and how geographic limits have changed throughout the past few years.

We can see how, for San Francisco, for example, the commercial and/or industrial areas are in proportion the lesser ones out of all of the categories, in terms of area, San Francisco's commercial and industrial area is the highest out of all categories.

We can also observe how the areas grow as the grade decreases, sometimes exchangably for D and C, but it is always the case that A and B grades have less total area.

```
agg_area = (
    holc.groupby(["city", "grade","grade_category"])["area_km2"]
        .sum()
        .reset_index()
)

fig_name_areas = "5_areas_per_grade.png"
fig, ax = plt.subplots(figsize=(12, 6))
```

```
sns.swarmplot(
    data=agg_area,
    x="city",
    y="area_km2",
    hue="grade_category",
    hue_order=grade_color_map.keys(),
    palette=grade_color_map,
    size=10
)

plt.title("Total Area (sq km) of HOLC -Graded Zones per City")
plt.xlabel("City")
plt.ylabel("Total Area (sq km)")
plt.legend(title="HOLC Grade",
           loc='lower center',
           bbox_to_anchor=(0.5, 1.15),
           ncol=5)

plt.tight_layout()
fig.savefig(fig_folder + fig_name_areas)
plt.show()
```